Learning Discriminative Representations of Landslides via Deep Metric Learning Under Limited Supervision

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***Abstract*—Detecting low-contrast targets from complex backgrounds in remote sensing, such as landslides, poses fundamental challenges for learning discriminative representations under limited expert annotations. Deep metric learning (DML) provides a principled framework for learning discriminative representations by explicitly shaping the embedding space to enhance intraclass compactness and interclass separability. However, most existing DML paradigms are developed for image-level settings, where entire images are treated as a single category, and fail to explicitly address the supervision requirements of instance-level object detection, because contrastive supervision is constrained to labeled foreground instances, lacking an explicit mechanism to leverage background regions. We propose OSG-UAHNM, a plug-and-play framework that learns discriminative landslide representations through loss augmentation. A batch-wise online sample generator (OSG) constructs instance-level contrastive pairs from both labeled instances and unlabeled background regions, while an uncertainty-aware hard negative miner (UAHNM) identifies high-quality hard negatives by jointly modeling classification confidence and localization precision. With negligible parameter overhead and zero inference cost, OSG-UAHNM can be integrated into diverse architectures, albeit with noticeable additional training computation. Experiments on a field-surveyed loess landslide dataset demonstrate consistent AP50:95 gains of 7.6, 6.5, 2.7, 1.1, and 1.8 percentage points on Faster R-CNN, YOLOv8n, YOLO11n, YOLO26n, and DETR, respectively. Cross-dataset validation on the publicly available Bijie landslide dataset further demonstrates the robustness of the framework across distinct geological settings and landslide scales. Embedding-space analysis shows that interclass separation increases by more than two orders of magnitude while maintaining intraclass compactness.**

***Index Terms*—Deep metric learning, loess landslide, object detection, representation learning.**

I. Introduction

REMOTE sensing object detection often requires identifying targets of low visual contrast against complex backgrounds [1], [2], with landslide detection as a typical case. As shown in Fig. 1 (top layer), postfailure landslide deposits gradually become vegetated, blurring boundaries and reducing visual contrast with surrounding terrain [3], [4]. This challenge is particularly acute on the Chinese Loess Plateau, where landslides constitute roughly one-third of China's annual geological disasters, and their frequency continues to rise under the influence of intensified human activities and extreme weather events [5], [6]. Intelligent landslide detection over the Loess Plateau faces two core constraints: limited training samples (costly expert annotation) and low target-background contrast, which severely impairs the learning of discriminative representations—key to distinguishing landslide instances from background clutter.

In recent years, AI-driven methods have made substantial progress in loess landslide detection, supporting automated disaster investigation. Existing methods mainly adopt an architecture-centric approach: they either directly use advanced detection networks for baseline evaluation or integrate specialized modules into existing detectors to create task-specific architectures that improve detection performance. For example, Ju et al. [7] benchmarked existing detectors and achieved 18.9% AP50:95 with Mask R-CNN; Yu et al. [8] improved YOLOX to 95.43% AP50 through architecture-specific modifications; Li et al. [9] proposed a two-stage detection-segmentation pipeline to enhance the accuracy of landslide localization; Liang et al. [10] introduced MAH-YOLO with a hybrid CNN-Transformer architecture, which further improved feature extraction for low-contrast targets.

Although these methods have improved performance in specific scenarios, they still exhibit clear limitations in terms of adaptability and robustness. Specifically, as shown in Fig. 1 (middle layer), such methods typically integrate advanced functional modules (such as enhanced decoupled detection heads

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Datasets and code are available at https://github.com/CHU-187cai/OSG-UAHNM.

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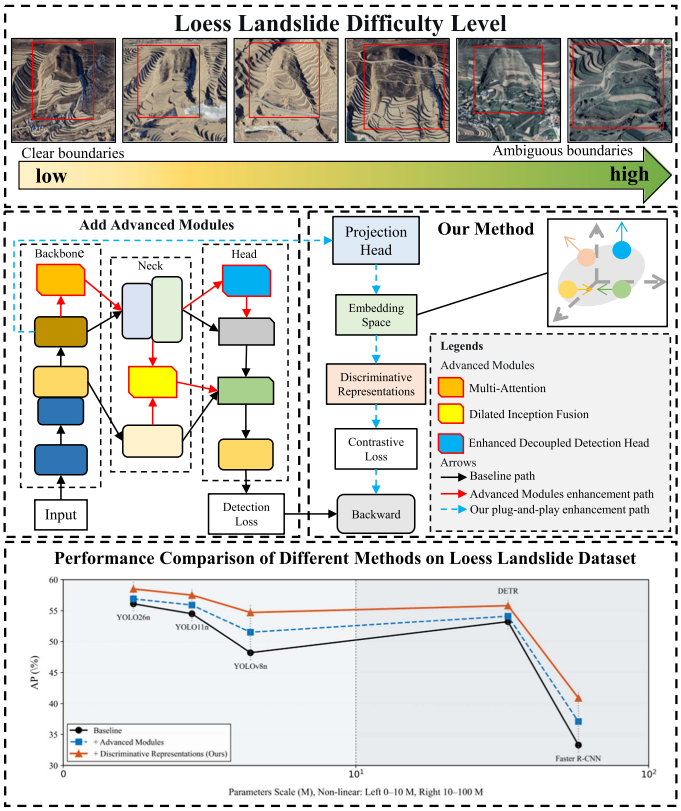


Fig. 1. Motivation of the proposed framework. Loess landslide detection is challenged by low target-background contrast, which becomes more severe as vegetation coverage increases. Conventional methods rely on architectural modifications (e.g., stacking advanced modules), which can saturate under limited annotations. Our method avoids modifying the detector structure and instead refines representation learning by adding a contrastive branch. High-level features from the backbone are fed into a projection head, where contrastive pairs are constructed via OSG and UAHNM. The contrastive loss is jointly optimized with the detection loss to improve feature discriminability.

[8], dilated Inception fusion modules [9], and multiattention modules [10]) into specific detectors in a tightly coupled manner. This architecture-specific design makes it difficult to transfer the improvements to other detector families—for example, the method developed for YOLOv8 [11] cannot be directly applied to Faster R-CNN [12] or DETR [13], which limits transferability across detector families.

In addition, these methods focus primarily on optimizing the model structure, neglecting improvements to representation learning, making it difficult to overcome the performance ceiling under conditions of limited labeled samples and low target-background contrast.

To address the above limitations, we focus on discriminative representation learning—an essential step for addressing the low contrast between loess landslides and background. Deep metric learning (DML) provides a principled framework for this task. As a powerful tool for learning discriminative representations, DML explicitly optimizes embedding spaces by enforcing distance-based similarity constraints [14], [15], providing a direct mechanism for addressing the low-contrast challenge. Closely related to DML [16] though sometimes treated as a distinct paradigm [15], contrastive learning has achieved remarkable success in both self-supervised pretraining (e.g., SimCLR [17], MoCo [18]) and supervised representation learning (e.g., SupCon [19]). Recent studies have further extended contrastive and metric learning paradigms to dense prediction tasks: FSCE [20] adopts contrastive proposal encoding for few-shot object detection, ICL [21] leverages contrastive learning for weakly supervised object detection, and PiCo [16] introduces pixel-level metric learning for semantic segmentation.

However, there is a clear gap between existing DML methods and the loess landslide detection task addressed here. Conventional DML relies on an image-level setting, in which entire images are assigned a single label as a whole rather than labeling individual instances within each image [17], [18], [19]. Although recent DML advances have been extended to dense prediction tasks [16], [20], [21], they are developed for natural scenes with clear semantic distinctions. Therefore, neither the image-level setting of traditional DML nor existing DML methods for dense prediction can adequately model the subtle relationship between landslide instances and their backgrounds.

To bridge this gap, we propose OSG-UAHNM, an instance-level DML framework. Unlike existing methods that mainly optimize the model structure [7], [8], [9], [10], our method enhances the discriminative capacity of feature representations at the algorithmic level. As illustrated in Fig. 1, conventional detectors rely on scaling structural complexity to boost performance, a strategy prone to saturation under limited labeled supervision. Our approach circumvents this bottleneck: without modifying the base detector architecture, we integrate an auxiliary deep metric learning branch to refine representation learning. This design yields consistent performance gains even under limited supervision, reducing the risk of saturation associated with structure-only optimization. Furthermore, since no structural modifications are required, the proposed framework maintains compatibility with diverse detectors, enabling plug-and-play deployment across different model families.

The main contributions are as follows.

1) We propose OSG-UAHNM, a plug-and-play DML framework that extends image-level metric learning to instance-level landslide detection, with negligible parameter overhead and zero inference cost, albeit with noticeable additional training computation.
2) OSG resolves the mismatch between image-level and instance-level learning by constructing instance-level contrastive pairs.
3) Uncertainty-aware hard negative miner (UAHNM) complements OSG by adaptively identifying informative hard negatives.

II. Related Work

A. Discriminative Representation in Remote Sensing

Discriminative representation is fundamental to visual-recognition performance [22]. It aims to capture the distinctive attributes of target objects while reducing interclass feature similarity, thereby facilitating effective category separation [23], [24]. This technique has achieved significant success in natural image analysis, largely owing to the clear object boundaries,

abundant labeled samples, and relatively stable semantic structures commonly found in natural images, which enable deep models to learn robust discriminative features more reliably. For example, PL-Net [23] leverages sufficient human image samples and clear body-part boundaries to guide the learning of discriminative representations for different body parts via part loss. Similarly, the unified deep network in [24] utilizes clear object boundaries and rich annotation information in natural images to promote category-wise feature separation in multilabel scenes through object parsing and a multilabel contrastive loss design.

In contrast to natural images, learning discriminative representations in remote sensing imagery is considerably more challenging because remote sensing scenes often exhibit severe spectral confusion between targets and surrounding backgrounds, as well as substantial scale and spatial heterogeneity, making effective target-background separation difficult [25], [26]. To address these issues, several studies have explored discriminative representation learning for remote sensing applications. For instance, Zhou et al. [25] proposed a dual-regularized stacked autoencoder (CDSAE), which incorporates local Fisher discriminant regularization and diversity regularization to enhance feature discriminability and alleviate spectral confusion. Zhao et al. [26] developed a co-enhanced global-part integration strategy that combines attention mechanisms with a collaborative enhancement loss to improve discriminative feature learning for remote sensing scene classification.

This challenge is particularly evident in loess landslide detection, where landslide regions often exhibit low contrast against complex backgrounds and require costly, expert annotation [1], [2]. However, although the aforementioned research demonstrates the potential of discriminative representation learning in remote sensing, such methods are designed for image-level classification and do not explicitly address the instance-level discriminative requirements of object detection. Therefore, a task-specific discriminative representation strategy is needed to improve the separability of landslide features from complex backgrounds and enhance detection accuracy.

B. Deep Metric Learning

DML is an established paradigm for learning discriminative representations by mapping samples into an embedding space where similarity is measured by distance [27]. Its core objective is to pull semantically similar samples closer while pushing dissimilar ones farther apart, thereby improving feature separability [28], [29]. Owing to its explicit distance-based supervision, DML is particularly attractive for tasks with limited labeled data, where highly discriminative representations are essential [30], [31]. To optimize the structure of the embedding space, a variety of loss functions have been developed [32], [33], including triplet loss [34], N-pair loss [35], and proxy-anchor loss [36], each enforcing discriminative similarity constraints from different optimization perspectives.

Closely related to DML, contrastive learning follows the same pull–push principle and has emerged as a powerful paradigm for representation learning [37]. The principal difference is the supervision source: conventional DML typically relies on label-defined positive and negative pairs, whereas contrastive learning often constructs them from different augmented views of the same sample in a self-supervised manner [15]. For example, CLSA [38] improves the learning of strongly augmented images by minimizing distribution divergence, while GraphACL [39] adopts an adversarial contrastive learning framework to optimize a negative sample bank and provide more informative negative samples. These studies demonstrate that unlabeled data can effectively support feature learning when appropriate similarity constraints are imposed.

Recently, metric and contrastive learning have increasingly been applied to dense prediction tasks beyond image-level recognition. FSCE [20] introduces contrastive learning into few-shot object detection, ICL [21] uses it for weakly supervised object detection, and PiCo [16] exploits pixel-level metric learning for semantic segmentation. These studies demonstrate the potential of embedding-based discriminative learning in detection and segmentation scenarios.

Another factor affecting DML performance is hard sample mining, especially hard negative mining. Since easy pairs contribute limited discriminative information, focusing on difficult positives and negatives can substantially improve embedding quality by sharpening decision boundaries. Representative studies such as HCM [40] and GGDHSCL [41] show that carefully selected hard negatives enhance feature discrimination and regularize the embedding distribution. In object detection, related techniques such as OHEM [42] and BoN [43] alleviate sample imbalance and improve training efficiency. Nevertheless, these methods mainly emphasize classification difficulty and sample reweighting, rather than explicitly structuring the embedding space from a metric learning perspective.

Despite these advances, existing DML methods remain poorly aligned with remote sensing object detection, especially loess landslide detection. Traditional DML methods are designed for image-level recognition, while existing instance-level DML approaches are mostly developed for natural images and place less emphasis on separating instances from complex backgrounds. Therefore, an instance-level DML framework tailored for separating landslides from complex backgrounds is needed to explicitly organize the embedding space.

III. Proposed Method

To address the aforementioned limitations, we propose OSG-UAHNM—a plug-and-play framework that learns discriminative landslide representations through loss augmentation. Before detailing the framework components, we formalize the problem and establish the mathematical foundation.

Let $\mathcal{I} = \{I_i\}_{i=1}^{N}$ denote a training set of N remote sensing images, where each image $I_i \in \mathbb{R}^{H\times W\times 3}$ contains M_i annotated landslide instances with bounding boxes $\mathcal{B}_i = \{b_{i,j}\}_{j=1}^{M_i}$, where $b_{i,j} = (x, y, w, h)$ specifies the spatial location and extent of the $\mathrm{j}-\mathrm{th}$ instance in image I_i. Given the limited supervision constraint ($N \ll 10^3$), our objective is to learn a feature encoder $f_\theta : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^d$ that maps image regions to a d-dimensional embedding space $\mathcal{Z}$, where landslide instances are tightly clustered while background regions are pushed far apart.

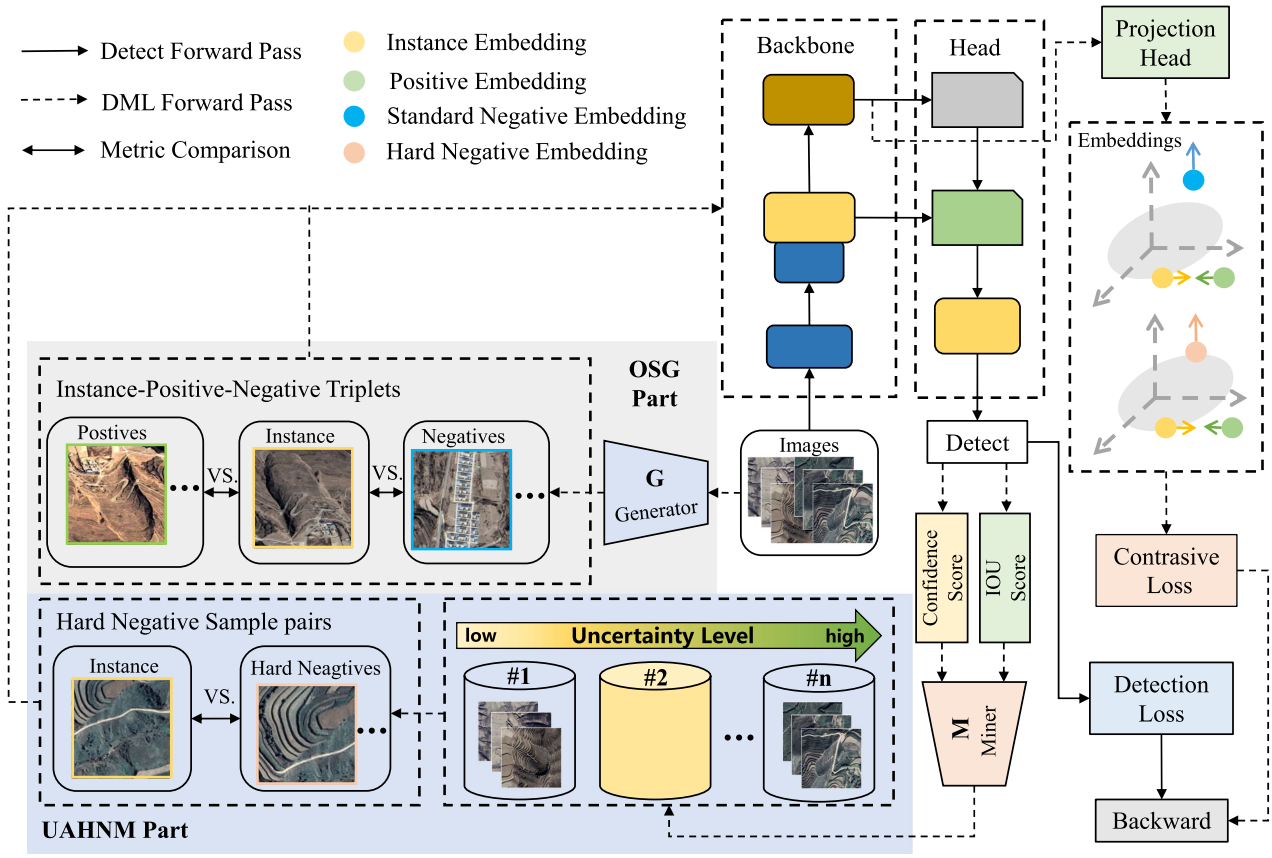


Fig. 2. Overview of the proposed framework. The OSG generates standard positive/negative triplets, while the UAHNM mines hard negatives based on uncertainty levels. Real images pass through the full detection pipeline, whereas generated samples reuse the detector's shared-weight backbone for feature extraction to reduce computational overhead. Instead of feeding these features into the detection head, they are passed to a projection head to produce embeddings, which are optimized via a weighted contrastive loss to learn discriminative landslide representations. Finally, the contrastive loss is combined with the detection loss for joint backward optimization.

As illustrated in Fig. 2, the OSG-UAHNM framework operates as an integrated system within the standard detection training loop. For each training image I_i with ground-truth boxes $\mathcal{B}_i$, the detector's backbone network f_θ performs forward propagation to generate feature maps, predicted bounding boxes $\hat{\mathcal{B}}_i = \{\hat{b}_{i,j}\}_{k=1}^{K_i}$ with confidence scores $\{s_{i,k}\}_{k=1}^{K_i}$, and predicted intersection-over-union (IoU) values $\{\mathrm{IOU}_{i,k}\}_{k=1}^{K_i}$ quantifying localization quality.

Simultaneously, the online sample generator (OSG) constructs instance-level contrastive pairs by exploiting both labeled instances and unlabeled background regions, while the UAHNM identifies high-quality hard negatives by jointly modeling classification confidence and localization uncertainty. These standard and hard sample pairs are processed through the shared-weight encoder to extract embeddings, which are then optimized via an uncertainty-aware structured contrastive loss (UASCL). The final training objective unifies detection and metric learning as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}\left(\hat{\mathcal{B}}_i, \mathcal{B}_i\right) + \lambda \cdot \mathcal{L}_{\text{contrastive}}\left(\mathcal{P}_i, \mathcal{H}_i; f_\theta\right) \quad (1)$$

where $\mathcal{L}_{\text{det}}$ represents the native detection loss, $\mathcal{P}_i$ and $\mathcal{H}_i$ denote the standard and hard sample pairs respectively, λ is a balancing hyperparameter, and the contrastive loss explicitly structures the embedding space to enhance interclass separation and intraclass compactness. Crucially, this framework introduces no architectural modifications—all enhancements are achieved through loss augmentation, making it architecture-agnostic and deployable across diverse detector families with negligible parameter overhead.

A. Online Sample Generator

The OSG module resolves the granularity mismatch between image-level metric learning and instance-level detection by dynamically constructing contrastive pairs directly from annotated instances within each training batch. As shown in Fig. 2, conventional contrastive frameworks operate on entire images, which is ill-suited for object detection where discrimination must occur at the level of individual instances. The OSG addresses this fundamental limitation by shifting from image-level to instance-level pair construction.

For each ground-truth bounding box $b_{i,j} \in \mathcal{B}_i$, we extract the corresponding image region $r_{i,j} = \mathrm{Crop}(I_i, b_{i,j}) \in \mathbb{R}^{H\times W\times 3}$ as a positive instance anchor. To form a positive pair that captures intraclass variation while maintaining semantic consistency, we apply two independent stochastic augmentation sequences T_1, T_2 sampled from a distribution $\mathcal{A}$ (including

random rotation, color jittering, and Gaussian blur) to this anchor: $P_{i,j}^{+} = T_1(r_{i,j})$ and $P_{i,j}^{+'} = T_2(r_{i,j})$. This produces two distinct yet semantically congruent views of the same landslide instance, establishing the foundation for intraclass compactness in the embedding space.

To ensure that standard negative samples are sufficiently discriminative from landslide instances, the OSG mines background regions by generating K random candidate boxes $\mathcal{C}_i = \{c_{i,k}\}_{k=1}^{K}$ uniformly distributed across the image and selecting only those with zero IoU with any ground-truth annotation. Although our dataset is generated with nonoverlapping tiles and split at the individual landslide-unit level, a bounding box is still an approximate representation of the landslide extent and may not fully cover all landslide-related features. To further reduce the risk of including landslide-related features outside the annotated bounding boxes, we extend each ground-truth box with a small safety buffer. For a box $b = [x_1, y_1, x_2, y_2]$, the buffered box is $b(+\epsilon) = [x_1 - \epsilon, y_1 - \epsilon, x_2 + \epsilon, y_2 + \epsilon]$, where ϵ is a small margin and coordinates are clipped to the image boundary. Based on this buffered box, we define the negative sample set as follows:

$$\mathcal{N}_i = \{c \in \mathcal{C}_i | \max_{b \in \mathcal{B}_i} \mathrm{IOU}\left(c, (b + \epsilon)\right) = 0\}. \tag{2}$$

This buffer-constrained zero-IoU sampling rule provides an additional safeguard against ambiguous background regions near landslide boundaries. If no candidate satisfies this condition, it is discarded. Each selected background patch $n \in \mathcal{N}_i$ is then resized to match the instance size, forming negative samples $P_{i,j}^{-} = \mathrm{Resize}(\mathrm{Crop}(I_i, n))$. Through this online process, the OSG supplies a continuous stream of instance-specific pairs $\mathcal{P}_i = \{(P_{i,j}^{+}, P_{i,j}^{+'}), (P_{i,j}^{+}, P_{i,j}^{-})\}$ that adapt dynamically to batch-specific characteristics during training.

All generated sample pairs are processed through the shared-weight backbone network f_θ for feature extraction and projection into a unified d-dimensional embedding space: $z_p = f_\theta(p)$ for all $p \in \mathcal{P}_i$, where $z_p \in \mathbb{R}^d$ represents the $\ell_2 -$ normalized embedding vector. This shared-weight design ensures that the same feature representations used for detection are simultaneously optimized for metric learning, creating a synergistic effect where improvements in embedding quality directly translate to enhanced detection performance. The OSG module thus establishes the necessary foundation for learning a discriminative embedding space without requiring external data or manual annotation of background regions.

B. Uncertainty-Aware Hard Negative Miner

While the OSG provides standard contrastive pairs that establish basic interclass separation, the most informative samples for refining decision boundaries are hard negatives—background regions that exhibit partial landslide characteristics and thus reside close to the positive cluster in embedding space. As illustrated in Fig. 2, the UAHNM systematically identifies these challenging samples by jointly modeling classification and localization uncertainties inherent in object detection. Unlike standard negatives that are easily separable, hard negatives force the model to learn subtle discriminative features that distinguish true landslides from visually similar background terrain.

Given a mini-batch of predicted bounding boxes $\hat{\mathcal{B}} = \{\hat{b}_k\}_{k=1}^{K}$ with confidence scores $\{s_k\}_{k=1}^{K}$ and ground truth boxes $\mathcal{B} = \{b_j\}_{j=1}^{M}$, the UAHNM first computes the maximum IoU between each prediction and all ground-truth boxes:$\mathrm{IOU}_k = \max_{b \in \mathcal{B}} \mathrm{IOU}(\hat{b}_k, b)$ for all k. Based on batch statistics $\mu_{\mathrm{IOU}} = \mathrm{mean}(\{\mathrm{IOU}_k\})$, $\sigma_{\mathrm{IOU}} = \mathrm{std}(\{\mathrm{IOU}_k\})$ (mean and standard deviation of IoU values) and $\mu_{\mathrm{s}} = \mathrm{mean}(\{\mathrm{s}_k\})$,$\sigma_{\mathrm{s}} = \mathrm{std}(\{\mathrm{s}_k\})$ (mean and standard deviation of confidence scores), a dynamic negative threshold τ_{neg} is established as follows:

$$\tau_{\mathrm{neg}} = \max\left(0.3, \mu_{\mathrm{IOU}} - 0.5\sigma_{\mathrm{IOU}}\right). \tag{3}$$

The threshold adapts to the current batch while maintaining a minimum value of 0.3 for training stability. This adaptation accounts for the fact that different training stages and batch compositions exhibit varying difficulty levels—early training produces more uncertain predictions requiring a lower threshold, while later stages benefit from stricter criteria.

Candidate negative samples are selected by filtering predictions that are poorly localized yet exhibit relatively high confidence:

$$\mathcal{C}_{\mathrm{hard}} = \left\{\hat{b}_k | \mathrm{IOU}_k \left\langle \tau_{\mathrm{neg}} \wedge s_k \right\rangle \mu_s - \sigma_s \right\}. \tag{4}$$

This selection strategy identifies samples that are characteristic of hard negatives: the model confidently predicts them as landslides (high s_k), yet they have poor spatial overlap with ground truth (low IOU_k), indicating semantic ambiguity. For each candidate $\hat{b}_k \in \mathcal{C}_{\mathrm{hard}}$, an uncertainty score u_k is computed by integrating two complementary components:

$$u_k = \frac{s_k}{\mu_s} \times \frac{\tau_{\mathrm{neg}} - \mathrm{IOU}_k}{\tau_{\mathrm{neg}}} \tag{5}$$

where the first term $\frac{s_k}{\mu_s}$ measures classification uncertainty (normalized prediction confidence relative to batch average) and the second term $\frac{\tau_{\mathrm{neg}} - \mathrm{IOU}_k}{\tau_{\mathrm{neg}}}$ quantifies localization uncertainty (relative gap between negative threshold and actual IoU). The multiplicative combination ensures that samples exhibiting both high classification uncertainty and poor localization receive elevated uncertainty scores, effectively capturing the dual nature of detection difficulty. A small constant $\epsilon = 10^{-6}$ can be added to denominators if needed to prevent numerical instability. All qualifying candidates are collected into the candidate set $\mathcal{C}$ along with their uncertainty scores into set $\mathcal{U}$.

Subsequently, a hard negative threshold τ_{hard} is established based on the distribution of uncertainty scores within the candidate set

$$\tau_{\mathrm{hard}} = \mu_u + 0.5\sigma_u \tag{6}$$

where $\mu_u = \mathrm{mean}(\mathcal{U})$ and $\sigma_u = \mathrm{std}(\mathcal{U})$ represent the mean and standard deviation of uncertainty scores within $\mathcal{C}_{\mathrm{hard}}$. This statistical approach automatically identifies outliers in the uncertainty distribution—samples that are significantly more challenging than the average candidate. To maintain training stability and focus on the most informative samples, the number of selected hard negatives $\mathcal{K}_{\mathrm{hard}}$ is constrained as $\mathcal{K}_{\mathrm{hard}} =$

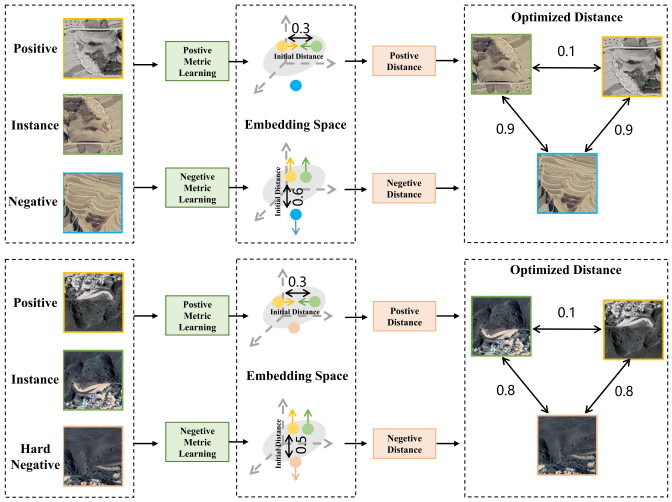


Fig. 3. Principle of metric learning with standard and hard negative pairs in our framework. Positive metric learning pulls positive pairs close, while negative metric learning pushes negative pairs apart. The circled points with arrows illustrate this constraint process in the embedding space. Hard negatives mined by UAHNM are kept closer to positives than standard negatives generated by OSG, with final separation degrees denoted by representative distance values. This expected embedding pattern is verified by the analysis of intersample distances in embedding space.

Algorithm 1: Uncertainty-Aware Hard Negative Miner.

Require: Predictions $\hat{\mathcal{B}} = \{\hat{b}_k\}_{k=1}^{K}$ with scores $\{s_k\}_{k=1}^{K}$
Require: Ground truths $\mathcal{B} = \{b_j\}_{j=1}^{M}$
Ensure: Hard negatives $\mathcal{H}$
$\mathrm{IoU}_k \leftarrow \max_{b\in\mathcal{B}} \mathrm{IoU}(\hat{b}_k, b), \forall k$
$\mu_{\mathrm{IoU}} \leftarrow \mathrm{mean}(\{\mathrm{IoU}_k\})$, $\sigma_{\mathrm{IoU}} \leftarrow \mathrm{std}(\{\mathrm{IoU}_k\})$
$\mu_{\mathrm{s}} \leftarrow \mathrm{mean}(\{\mathrm{s}_k\})$, $\sigma_s \leftarrow \mathrm{std}(\{\mathrm{s}_k\})$
$\tau_{neg} \leftarrow \max(0.3,\ \mu_{\mathrm{IoU}} - 0.5\sigma_{\mathrm{IoU}})$
$\mathcal{C} \leftarrow \emptyset, \mathcal{U} \leftarrow \emptyset$
for $k = 1$ to K **do**
  if $\mathrm{IoU}_k < \tau_{\mathrm{neg}}$ **and** $s_k > \mu_s - \sigma_{\mathrm{s}}$ **then**
    $u_k = \frac{s_k}{\mu_{\mathrm{s}}} \times \frac{\tau_{\mathrm{neg}} - \mathrm{IoU}_k}{\tau_{\mathrm{neg}}}$ (add $\epsilon = 10^{-6}$ if needed)
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\hat{b}_k,\ s_k,\ u_k)\}$
    $\mathcal{U} \leftarrow \mathcal{U} \cup \{u_k\}$
  end if
end for
$\mu_u \leftarrow \mathrm{mean}(\mathcal{U})$, $\sigma_u \leftarrow \mathrm{std}(\mathcal{U})$
$\tau_{\mathrm{hard}} \leftarrow \mu_u + 0.5\sigma_u$
$\mathcal{H}' \leftarrow \{(\hat{b}_k, s_k \in \mathcal{C}) | u_k \rangle \tau_{hard}\}$
$\mathcal{K}_{\mathrm{hard}} \leftarrow \lceil 0.3 \times |\mathcal{C}| \rceil$
$\mathcal{H} \leftarrow \mathrm{ToP} - K_{\mathrm{hard}}(\mathcal{H}', u_k)$
return $\mathcal{H}$

$\lceil 0.3 \cdot |C| \rceil$. Preliminary hard negatives are first filtered as $\mathcal{H}' = \{(\hat{b}_k, s_k \in \mathcal{C}) | u_k \rangle \tau_{\mathrm{hard}}\}$, and the final hard negatives $\mathcal{H} \subset \mathcal{C}_{\mathrm{hard}}$ are obtained by selecting the $\mathrm{TOP} - K_{\mathrm{hard}}$ candidates with the highest uncertainty scores

$$\mathcal{H} \leftarrow \mathrm{TOP} - K_{\mathrm{hard}}\left(\mathcal{H}', u_k\right). \tag{7}$$

As visualized in the uncertainty level ranking shown in Fig. 2., these hard negatives are organized into bins (#1, #2, …, #n) according to their uncertainty levels. The proposed UAHNM automatically identifies the appropriate difficulty bin from the current batch and selects hard negatives within this range to improve learning efficiency. As illustrated in Fig. 3, the resulting hard negatives remain closer to positives than standard negatives in the embedding space, guiding the metric learning process. The method operates entirely on batch statistics without requiring manual hyperparameter tuning, jointly models classification and localization uncertainties through a unified scoring mechanism, adapts dynamically to different training stages and batch characteristics, and maintains computational efficiency by utilizing only existing forward propagation outputs.

C. Uncertainty-Aware Structured Contrastive Loss

To learn discriminative embedding representations that explicitly enhance interclass separation and intraclass compactness, we propose the UASCL, which simultaneously models both classification and localization uncertainties within the embedding space through a compact mathematical formulation.

For a sample pair (p_a, p_b) with embeddings $z_a = f_\theta\left(p_a\right)$ and $z_b = f_\theta\left(p_b\right)$, the UASCL is defined as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{UASCL}}\left(z_a, z_b\right) &= y\left(1 + \beta \cdot u_a^{\mathrm{cls}}\right) d(z_a, z_b)^2 \\ &+ (1-y)\left(1 + \alpha \cdot u_a\right) \max(0, m - d\left(z_a, z_b\right))^2\end{aligned} \tag{8}$$

where $y \in \{0, 1\}$ indicates the sample label (0 for negative, 1 for positive), $d(z_a, z_b) = 1 - \frac{z_a{}^T z_b}{\|z_a\| \|z_b\|}$ is the cosine distance in embedding space (hence $d \in [0, 2]$), α and β are hyperparameters controlling the emphasis on hard negative and hard positive samples respectively, u_a is the comprehensive uncertainty score for hard negatives, u_a^{cls} is the classification uncertainty for positive samples, and m is an adaptive margin (AM) dynamically adjusted based on batch characteristics.

The UASCL incorporates three adaptive components. For positive pairs ($y = 1$), the weight $(1 + \beta \cdot u_a^{\mathrm{cls}})$ increases when the model shows insufficient confidence on landslide regions, compelling focus on hard positives. For negative samples ($y = 0$), the weight $(1 + \alpha \cdot u_a)$ increases with joint classification-localization uncertainty, enhancing separation of hard negatives. The AM $m = m_0 + \gamma_1 \cdot \mu_s - \gamma_2 \cdot \sigma_{\mathrm{IOU}}$ enlarges under high batch confidence to boost discriminability, and shrinks under localization instability to avoid overpenalization.

The UASCL is averaged over all valid sample pairs in a batch:

$$\mathcal{L}_{\mathrm{contrastive}} = \frac{1}{|\mathcal{P} \cup \mathcal{H}|} \sum_{(z_a, z_b) \in \mathcal{P} \cup \mathcal{H}} \mathcal{L}_{\mathrm{UASCL}}\left(z_a, z_b\right) \tag{9}$$

and integrated with the baseline detector's native loss. For Faster R-CNN, the native loss combines the region proposal network (RPN) loss and detection head loss:

$$\mathcal{L}_{\mathrm{det}}^{\mathrm{FRCNN}} = \mathcal{L}_{\mathrm{cls}}^{\mathrm{RPN}} + \mathcal{L}_{\mathrm{reg}}^{\mathrm{RPN}} + \mathcal{L}_{\mathrm{cls}}^{\mathrm{det}} + \mathcal{L}_{\mathrm{reg}}^{\mathrm{det}} \tag{10}$$

where $\mathcal{L}_{\mathrm{cls}}^{\mathrm{RPN}}$ and $\mathcal{L}_{\mathrm{reg}}^{\mathrm{RPN}}$ denote the classification and regression losses in the RPN stage, while $\mathcal{L}_{\mathrm{cls}}^{\mathrm{det}}$ and $\mathcal{L}_{\mathrm{reg}}^{\mathrm{det}}$ represent the classification and regression losses in the detection head.

For the YOLO series (v8, 11, 26), the loss typically comprises bounding box regression, object confidence, and classification

losses

$$\mathcal{L}_{\text{det}}^{\text{YOLO}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{dfl}} \tag{11}$$

where $\mathcal{L}_{\text{box}}$ denotes the CIoU loss for bounding box regression, $\mathcal{L}_{cls}$ is the classification loss (binary cross-entropy), $\mathcal{L}_{\text{dfl}}$ specifically represents the distribution focal loss in anchor-free YOLOv8 and v11(note that YOLO26 removes this component for simplified architecture), and λ_{box}, λ_{cls}, λ_{dfl} are balancing weights. For DETR, an end-to-end Transformer-based detector utilizing bipartite matching via the Hungarian algorithm, the loss is as follows:

$$\mathcal{L}_{det}^{DETR} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{bbox}}\mathcal{L}_{\text{bbox}} + \lambda_{\text{giou}}\mathcal{L}_{\text{giou}} \tag{12}$$

where $\mathcal{L}_{\text{cls}}$ is the focal loss for class prediction, $\mathcal{L}_{\text{bbox}}$ and $\mathcal{L}_{\text{giou}}$ are the ℓ_1 loss and GIOU loss between predicted and target bounding boxes, respectively.

The unified joint optimization objective integrating detection and metric learning is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \delta \cdot \mathcal{L}_{\text{contrastive}} \tag{13}$$

where $\mathcal{L}_{\text{det}}$ represents the native loss of any baseline detector ($\mathcal{L}_{\text{det}}^{\text{FRCNN}}$, $\mathcal{L}_{\text{det}}^{\text{YOLO}}$, or $\mathcal{L}_{\text{det}}^{\text{DETR}}$), and δ is a tunable weighting coefficient that balances the importance of the two

loss terms.

By combining the UASCL loss function with the UAHNM hard negative mining method, the two components provide a unified procedure for learning from informative samples: the UAHNM identifies high-uncertainty samples during forward propagation, while UASCL applies stronger optimization signals to these samples during backward propagation.

D. Implementation Details and Plug-and-Play Integration

The proposed OSG-UAHNM is a lightweight plug-and-play module. Integration is achieved solely by inserting the module into the standard training loop of each detector family and adding its output (contrastive loss) to the native detection loss with a balancing weight δ.

For two-stage (Faster R-CNN) and transformer-based (DETR) detectors, the module is invoked once per iteration before the native loss computation. For single-stage YOLO variants, it is integrated within the loss class by wrapping the deepest backbone stage. In all cases, the module reuses backbone features via a lightweight SmartFeatureExtractor wrapper, ensuring detection representations are directly co-optimized by the metric learning objective.

This introduces approximately 0.012 M extra parameters across all architectures. Relative to the original model sizes, the overhead is 0.02% for Faster R-CNN and DETR (both 60 M parameters), and 0.38%–0.50% for the YOLO variants (2.4–3.2 M parameters). At inference time, the module is fully disabled, incurring zero extra parameters or floating point operations (FLOPs).

Table I quantitatively evaluates the computational overhead induced by the auxiliary contrastive branch under a fixed input resolution of 640 × 640, including inference FPS, single-epoch training time, and peak GPU memory usage. Models marked with an asterisk denote baselines integrated with OSG-UAHNM. The statistics reveal that the proposed module has a negligible effect on inference speed and requires limited additional memory. The longer training time reflects the additional positive and hard-negative contrastive pairs processed in each iteration, and this additional training cost is accompanied by consistent gains in landslide-detection accuracy.

TABLE I
COMPUTATIONAL OVERHEAD COMPARISON ON BASELINE DETECTORS WITH AND WITHOUT OSG-UAHNM

Model	Input	FPS	Train Time (s/epoch)	Peak GPU Mem (GB)
Faster R-CNN	640	17.59	57.19	6.65
Faster RCNN*	640	17.66	111.31	6.70
YOLOv8n	640	151.05	3.11	1.93
YOLOv8n*	640	151.04	9.54	1.93
YOLO11n	640	124.66	3.29	2.15
YOLO11n*	640	124.23	9.56	2.18
YOLO26n	640	120.44	4.12	2.37
YOLO26n*	640	120.91	12.50	2.54
DETR	640	51.67	45.93	6.04
DETR*	640	51.36	74.05	6.09

TABLE II
AVERAGE COUNT OF CONTRASTIVE SAMPLES SAMPLED PER TRAINING BATCH

Model	BS	Batches	Avg Pos	Avg Neg	Avg Hard
Faster R-CNN*	4	5104	3.98	3.98	0.48
YOLOv8n*	16	2900	15.66	15.66	2.34
YOLO11n*	16	2900	15.66	15.66	2.21
YOLO26n*	16	5800	15.66	15.66	2.15
DETR*	4	5060	4.00	4.00	0.76

Table II records the average count of positive samples, standard negative samples and hard negative samples generated within each training batch for all detectors. The statistics verify that the OSG consistently produces balanced positive-negative pairs throughout training, and the UAHNM adaptively selects informative ambiguous candidates for all detection architectures. The total number of positive and negative pairs fluctuates under the early-stopping strategy, and the number of hard negatives is further limited by the warm-up schedule, which is uniformly set to 25 epochs in all experiments.

Faster R-CNN, the YOLO series, and DETR differ architecturally in candidate extraction, hard negative mining, feature selection and patch preprocessing, with the principal configuration differences summarized in Table III. NMS with IoU = 0.5 filters predictions before UAHNM sampling for Faster R-CNN and YOLO, while DETR directly uses raw decoder queries without NMS filtering. We adopt distinct feature extraction strategies tailored to each model's native design: Faster R-CNN uses only Layer-4 deep features because this stage holds the

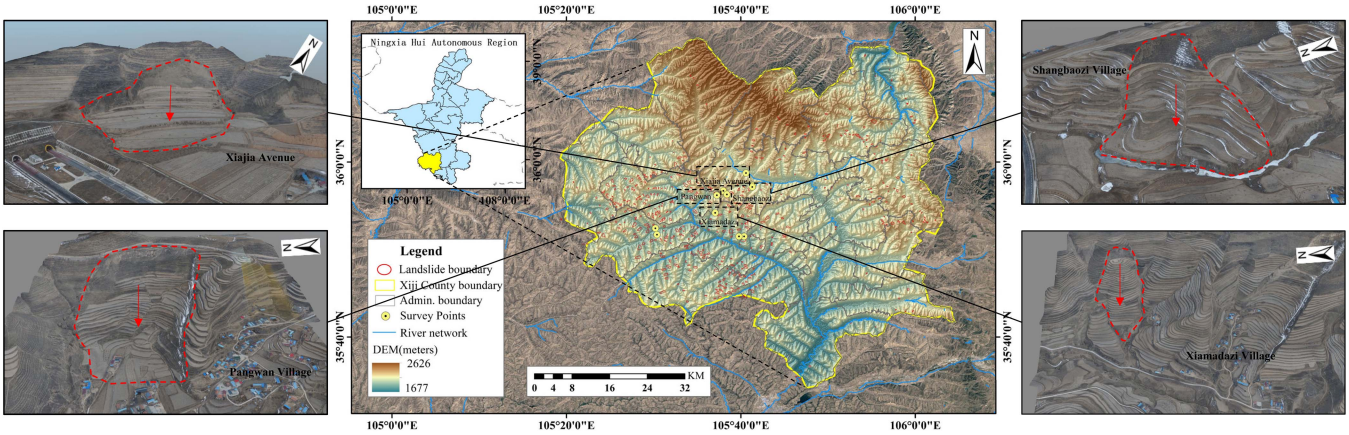


Fig. 4. Overview of the study area and representative landslides. The central thematic map shows the study area (Xiji County, Ningxia Hui Autonomous Region) and the distribution of landslides. The four surrounding panels display tiled 3-D visualizations of representative landslide sites derived from UAV photogrammetry, namely the Xia Jia Avenue, Shangbaozi Village, Pangwan Village, and Xiamadazi Village landslides.

TABLE III
DEDICATED CONFIGURATION SETTINGS FOR THREE DETECTOR ARCHITECTURES

Model	NMS	Feature Layers	Crop Size	Projection Head
Faster R-CNN	✓	Layer 4	256 × 256	GAP + Flatten
YOLO Series	✓	P3/P4/P5	256 × 256	GAP + Flatten
DETR	×	All Layers	256 × 256	GAP + Flatten

richest high-level semantic information to distinguish landslides from cluttered terrain backgrounds; YOLO leverages P3/P4/P5 multiscale pyramids to cover landslides with highly variable sizes in remote sensing imagery. Because DETR lacks built-in multiscale fusion, we concatenate features from all backbone stages. A unified 256 × 256 crop size and identical global average pooling (GAP) + Flatten projection head are applied across all models to eliminate confounding variables and support fair comparisons.

IV. EXPERIMENTS

A. Dataset

This study focuses on loess landslides because their characteristically ambiguous boundaries and high spectral similarity to the surrounding terrain present a substantial object-detection challenge and provide an appropriate testbed for the framework. Given the general scarcity of reliably validated, ground-truthed landslide annotations, we conducted a field survey in Xiji County in March 2025 to manually verify and delineate the landslides, thereby improving the accuracy of the experimental labels. Fig. 4 provides an overview of our study area, including field photographs captured during the survey, corresponding high-resolution satellite imagery from Google Earth Pro, and UAV-based photogrammetric models of representative landslides.

Our study area lies within Xiji County, situated in a typical loess hilly-gully region at 1500–2200 m elevation and within the meizoseismal zone of the 1920 M8.5 Haiyuan earthquake [44], where seismic faulting and long-term erosion have produced extensive historical landslide groups [45]. We manually identified and delineated 511 loess landslides using high-resolution orthoimagery from Google Earth Pro. To be compatible with our deep metric learning approach, all landslide images were uniformly cropped to 640×640 pixels, with each image containing three spectral channels (RGB).

The publicly available Bijie landslide dataset [48] is used for cross-dataset validation of robustness and supplementary scale-wise performance analysis. Covering landslides across a wide range of spatial scales, this dataset contains 770 annotated samples, all processed into identical 640 × 640 RGB tiles to maintain experimental consistency with the loess landslide dataset. All tiles were generated using nonoverlapping sliding-window cropping, with no overlapping geographic areas or duplicate samples. The training, validation, and test subsets were organized at the individual-landslide level so that each landslide body and its surrounding terrain appeared in only one subset, thereby preventing data leakage.

B. Settings and Evaluation Metrics

All experiments in this study were conducted on a Windows workstation equipped with an NVIDIA GeForce RTX 4060 GPU (16 GB VRAM) and an Intel Core i5-12400F CPU. All models were trained for 100 epochs using their respective default training configurations. The YOLO series models (YOLOv8, YOLO11, YOLO26) used a batch size of 16, while Faster R-CNN and DETR used a batch size of 4. The Xiji loess landslide dataset was randomly divided into training, validation and test subsets at a ratio of 7:1.5:1.5, and the Bijie landslide dataset was divided at a ratio of 6:2:2, providing sufficient data for hyperparameter optimization and final quantitative evaluation.

We employ standard object detection metrics including Precision (P), Recall (R), F1-score, average precision (AP), and mean

average precision (mAP), calculated as shown in (14)–(18). Since our dataset contains only a single landslide category, AP and mAP are equivalent in this study. For Faster R-CNN and YOLO series models, we report AP50:95 (averaged over IoU thresholds from 0.5 to 0.95), AP50, precision, recall, and F1-score (at IoU threshold of 0.5). For DETR, we adopt the standard COCO evaluation suite, reporting AP, AP50, AP75, and metrics for large objects (APL and ARL).

Official coco benchmarks separate targets by area thresholds 32^2 and 96^2 into small AP_S, medium AP_M, and large AP_L groups. Under this standard division, Xiji loess landslides fall almost entirely into the large category, whereas the Bijie dataset lacks sufficient small landslide instances. We therefore redefine custom area boundaries at 141^2 and 224^2 and derive corresponding metrics AP_S*, AP_M*, AP_L*. Even with revised area thresholds, loess landslides from our primary study area remain predominantly large-scale geohazards, which reflects the intrinsic geological trait that loess failures form uniformly extensive sliding masses. The balanced size distribution of landslides within the Bijie dataset makes it a more appropriate testbed to analyze performance discrepancies of our method across varying landslide scales.

We prioritize AP50:95 as the primary benchmark because single-threshold metrics (AP50 and F1) can be overly lenient for large-scale loess landslides. The expansive bounding boxes make achieving a 0.5 IoU relatively easy, while AP50:95 provides a more rigorous measure of localization accuracy across stricter thresholds. Unless otherwise specified, AP in this study refers to AP50:95

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

$$F1 - \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

$$\text{AP} = \int_0^1 \text{Precision}\,(\text{Recall})\, d\,(\text{Recall}) \tag{17}$$

$$\text{mAP} = \sum_{i=0}^{n} \frac{1}{n} \int_0^1 \text{Precision}\,(\text{Recall})\, d\,(\text{Recall})\,. \tag{18}$$

In the above equations, TP, FP, and FN denote True Positives, False Positives, and False Negatives, respectively.

C. Baseline Models for Comparison

As shown in Table IV, to select baseline models for a fair comparison with our proposed method, we systematically evaluated various backbone networks and architectural configurations on the loess landslide dataset, considering AP, parameter count, and FLOPs. Given the significant architectural evolution within the YOLO series and the need for comprehensive evaluation across different detector paradigms, we selected YOLOv8n, YOLO11n, and YOLO26n as representative one-stage detectors. For the two-stage paradigm, we chose Faster R-CNN with ResNet-101 backbone, and for the Transformer-based end-to-end detection paradigm, we selected DETR with ResNet-101 backbone. This selection covers mainstream detection architectures while maintaining computational feasibility.

TABLE IV
PERFORMANCE AND COMPLEXITY COMPARISON OF BASELINE MODELS ON THE LOESS LANDSLIDE DATASET (VALIDATION SET)

Model	AP(%)	Parameters(M)	FLOPs(G)
Faster RCNN	37.8	60.0	181.0
YOLOv8n	51.7	3.2	8.7
YOLO11n	56.8	2.6	6.5
YOLO26n	57.9	2.4	5.4
DETR	53.9	60.0	97.0

TABLE V
PERFORMANCE COMPARISON OF VARIOUS MODELS ON THE LOESS LANDSLIDE DATASET (TEST SET)

Model	AP (%)	AP_{50} (%)	P (%)	R (%)	F1 (%)
Faster R-CNN	33.3	89.7	80.5	90.9	85.4
Faster R-CNN*	**40.9**	**93.6**	**94.7**	**93.5**	**94.1**
YOLOv8n	48.2	**98.0**	89.9	92.7	91.3
YOLOv8n*	**54.7**	97.2	**93.7**	**96.1**	**94.9**
YOLO11n	54.5	97.9	94.3	91.3	92.8
YOLO11n*	**57.2**	**98.2**	**94.9**	**98.7**	**96.7**
YOLO26n	56.1	96.8	**94.3**	94.8	**94.5**
YOLO26n*	**57.2**	**97.7**	91.4	**96.8**	94.0

The bold values represent superior performance achieved by our method (marked with *) compared against baseline approaches.

TABLE VI
COCO EVALUATION OF DETR ON THE LOESS LANDSLIDE DATASET (TEST SET)

Model	AP (%)	AP_{50} (%)	AP_{75} (%)	AP_L (%)	AR_L (%)
DETR	53.2	95.0	53.3	53.2	80.4
DETR*	**55.0**	**98.9**	**55.3**	**55.0**	**83.6**

D. Overall Performance Comparison

Tables V and VI compare the baseline models and their OSG-UAHNM variants on the test set, where an asterisk (∗) following the model name indicates the incorporation of our proposed OSG-UAHNM framework. Table V reports metrics for Faster R-CNN and YOLO series models, while Table VI focuses on the DETR model using the COCO evaluation suite. As mentioned previously, we use AP50:95 as the core metric for accuracy assessment.

Table V shows that integrating OSG-UAHNM increased AP for Faster R-CNN, YOLOv8n, YOLO11n, and YOLO26n by 7.6, 6.5, 2.7, and 1.1 percentage points, respectively. Given that AP50:95 is a comprehensive and stringent metric considering multiple IoU thresholds and recall levels, these improvements indicate that the method enhances the discriminative capability

TABLE VII
PERFORMANCE COMPARISON OF VARIOUS MODELS ON THE BIJIE LANDSLIDE DATASET (TEST SET)

Model	AP (%)	AP_{50} (%)	P (%)	R (%)	F1 (%)
Faster R-CNN	54.4	97.1	95.4	94.8	95.1
Faster R-CNN*	**59.6**	**98.3**	**97.4**	**96.1**	**96.7**
YOLOv8n	59.3	97.3	**94.2**	94.8	**94.5**
YOLOv8n*	**62.2**	**97.8**	93.0	**94.8**	93.9
YOLO11n	62.8	99.3	96.7	94.8	95.7
YOLO11n*	**65.0**	**99.4**	**97.4**	**98.1**	**97.7**
YOLO26n	60.9	95.7	92.8	92.9	92.8
YOLO26n*	**63.5**	**96.5**	**94.4**	**95.0**	**94.7**

The bold values represent superior performance achieved by our method (marked with *) compared against baseline approaches.

TABLE VIII
COCO EVALUATION OF DETR ON THE BIJIE LANDSLIDE DATASET (TEST SET)

Model	AP (%)	AP_{50} (%)	AP_{75} (%)	AR_L (%)
DETR	62.4	95.2	72.4	83.5
DETR*	**65.4**	**95.9**	**78.6**	**84.0**

TABLE IX
SCALE-WISE AP PERFORMANCE OF ALL DETECTORS ON THE BIJIE LANDSLIDE DATASET (TEST SET)

Model	AP_M (%)	AP_L (%)	APs* (%)	AP_M* (%)	AP_L* (%)
Faster R-CNN	52.0	54.9	50.6	57.9	59.1
Faster R-CNN*	**53.6**	**60.6**	**57.6**	**61.9**	**60.2**
YOLOv8n	49.0	60.5	56.8	61.4	61.7
YOLOv8n*	**50.5**	**63.2**	**58.0**	**62.4**	**67.6**
YOLO11n	56.2	63.5	56.3	64.7	66.7
YOLO11n*	**62.1**	**65.3**	**59.1**	**65.8**	**71.2**
YOLO26n	50.8	62.0	53.2	62.9	67.7
YOLO26n*	**53.8**	**65.1**	**56.3**	**65.1**	**70.9**
DETR	60.1	62.3	61.8	64.9	61.2
DETR*	**62.5**	**65.9**	**63.3**	**72.0**	**64.6**

The bold values represent superior performance achieved by our method (marked with *) compared against baseline approaches.

of the models. The largest gain was observed for Faster R-CNN, which had the lowest baseline performance. This result indicates that lower-performing detectors can benefit from richer instance-level metric-learning supervision. For DETR (see Table VI), our method achieved an AP improvement of 1.8 percentage points along with consistent gains across AP50, AP75, and APL metrics, indicating compatibility with Transformer-based architectures.

Beyond AP improvements, our method also improves other metrics. For instance, Faster R-CNN$*$ achieved increases of 14.2 and 2.6 percentage points in precision and recall, respectively, relative to its baseline, resulting in an 8.7-percentage-point improvement in F1-score. Similarly, YOLOv8n$*$ and YOLO11n$*$ demonstrated balanced improvements in both precision and recall, with F1-scores increasing by 3.6 and 3.9 percentage points, respectively. These results indicate improvements in both localization accuracy and overall detection performance.

To evaluate cross-dataset robustness, we conducted parallel experiments on the Bijie landslide dataset. Tables VII and VIII show consistent AP improvements across all detector architectures and both landslide-detection settings.

Table IX further dissects performance across small, medium, and large landslide sizes using our customized AP$*$ indicators. For all baseline detectors, AP values follow the expected order of large $>$ medium $>$ small. After integrating OSG-UAHNM, all three size groups achieve consistent gains, averaging 3.1, 3.1, and 3.6 percentage points for small, medium, and large landslides, respectively. The largest gain on large landslides may reflect their intact morphology and rich internal textural details, which facilitate wider feature margins in the contrastive embedding space. Meanwhile, the stable gains on small and medium landslides demonstrate that the proposed metric learning supervision can effectively enhance feature discriminability even for targets with limited spatial information.

Despite consistent improvements on the core AP50:95 metric, minor performance tradeoffs on individual single-threshold metrics are observed across both datasets. On the loess landslide dataset, YOLOv8n experiences a slight AP50 drop from 98.0% to 97.2%, and YOLO26n shows marginal declines in both precision and F1-score. On the Bijie landslide dataset, YOLOv8n shows drops of 1.2 and 0.6 percentage points in precision and F1-score, respectively, relative to its baseline. These tradeoffs may be associated with the optimization objective of contrastive embedding learning: the contrastive loss encourages the model to reduce overconfident predictions on easily distinguishable samples, to expand feature margins between ambiguous hard negatives and background terrain. Although this adjustment slightly degrades performance under the loose IoU $= 0.5$ threshold, it delivers notable gains on the core multi-IoU AP50:95 metric, which better reflects practical localization accuracy for landslide detection. These results indicate a tradeoff between relaxed single threshold metrics and stricter multi-IoU metrics.

Fig. 5 provides qualitative comparisons between baseline models and their enhanced counterparts. Fig. 6 provides additional qualitative detection comparisons covering landslide targets of different sizes from the Bijie dataset. The visualization shows that models integrated with our OSG-UAHNM framework produce more accurate bounding boxes with higher confidence scores, particularly in challenging scenarios involving ambiguous boundaries and high background similarity—precisely the conditions for which metric learning is intended to improve discrimination.



Fig. 5. Comparison of landslide detection results on the loess landslide dataset (test set). Each row represents a test sample, with columns showing results from different detection methods. Red boxes indicate predictions with confidence scores, and green boxes show ground truth. Methods with asterisk (∗) are enhanced with the proposed framework, demonstrating improved detection performance across various baseline detectors.

E. Related Method Comparison

We compare the proposed OSG-UAHNM framework with four representative landslide-detection methods spanning two technical approaches—metric learning and hard example mining—in comparative experiments. FSCE [20] introduces a contrastive proposal encoding branch that directly optimizes the embeddings of RoI aligned region proposals, sharing the same embedding-optimization principle as this work. DeFRCN [49] adopts a decoupled detection architecture with a cosine-similarity-based classification head. Although originally developed for few-shot scenarios, its cosine similarity measurement mechanism is methodologically similar to our metric learning design. OHEM [42] is a classic online hard example mining algorithm that dynamically selects high-loss samples to strengthen the model's learning of ambiguous targets. PISA [50] is an advanced sample importance weighting method that boosts detection performance by elevating the training weight of informative samples that contribute strongly to accuracy.

All four comparison methods were originally proposed and validated on the Faster R-CNN detection paradigm. To control for backbone-related differences, we reimplemented each method following the algorithmic details in its original publication, and uniformly adopted the Faster R-CNN detector with a ResNet-101-FPN backbone. All experiments were conducted under identical configurations of dataset split, input resolution, and training hyperparameters.

TABLE X
PERFORMANCE COMPARISON WITH RELATED METHODS ON THE XIJI LOESS LANDSLIDE DATASET (TEST SET)

Method	AP(%)	F1 (%)
Baseline	33.3	85.4
FSCE	36.8	89.2
DeFRCN	35.2	87.5
OHEM	38.9	92.4
PISA	39.5	92.7
OSG-UAHNM(ours)	**40.9**	**94.1**

Quantitative comparison results on the Xiji loess landslide dataset and the Bijie landslide dataset are summarized in Table X and Table XI, respectively, with corresponding visual comparisons provided in Figs. 7 and 8. Overall, the proposed OSG-UAHNM achieves the highest AP and F1-score on both datasets, delivering the most balanced and robust optimization effect. On the Xiji loess landslide dataset with low visual contrast and high detection difficulty, our method yields an AP improvement of 7.6 percentage points relative to the baseline, significantly outperforming all other compared methods. As an advanced hard sample mining approach, PISA also delivers competitive

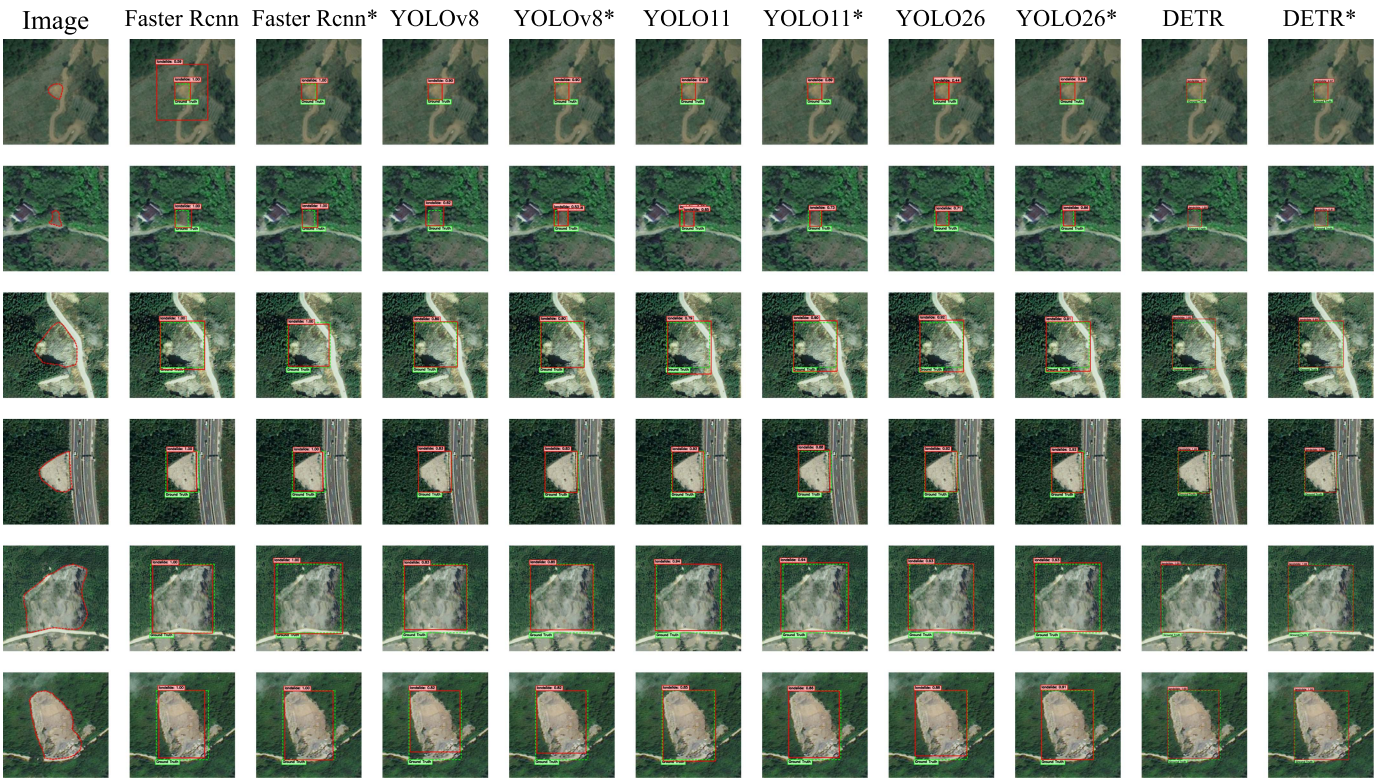


Fig. 6. Comparison of landslide detection results on the Bijie landslide dataset (test set). From top to bottom, the landslide examples are grouped in pairs, corresponding respectively to small-, medium-, and large-scale landslides according to our custom size categorization.

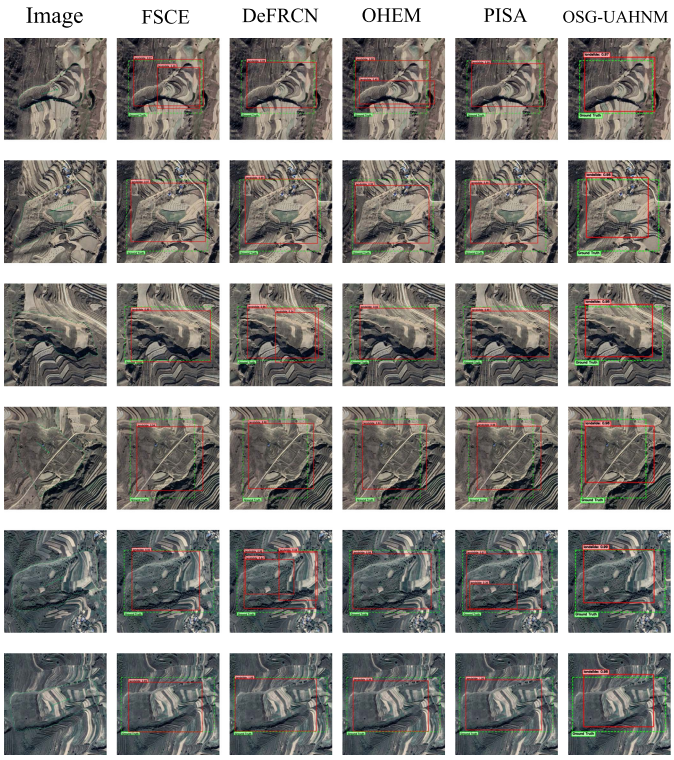


Fig. 7. Performance comparison of different methods on the Xiji loess landslide dataset (test set).

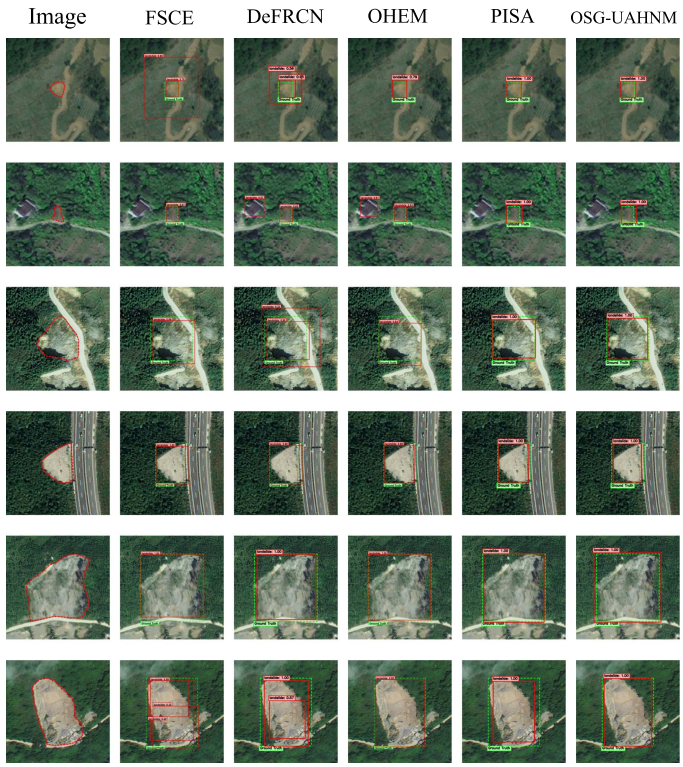


Fig. 8. Performance comparison of different methods on the Bijie landslide dataset (test set).

TABLE XI
PERFORMANCE COMPARISON WITH RELATED METHODS ON THE BIJIE LANDSLIDE DATASET (TEST SET)

Method	AP(%)	F1 (%)
Baseline	54.4	95.1
FSCE	56.2	95.4
DeFRCN	55.4	94.3
OHEM	56.8	96.2
PISA	57.1	95.5
OSG-UAHNM(ours)	**59.6**	**96.7**

performance, but still suffers from partial misclassification on samples with extremely low visual contrast.

Cross-dataset comparison reveals distinct adaptation patterns across method classes. For hard example mining methods, their performance gains are more strongly attenuated on the Bijie dataset with higher visual contrast. This attenuation may reflect the fact that such methods only adjust the training weights of samples without directly enhancing the feature separability between landslides and background in the embedding space. In high-contrast scenes where positive and negative samples are inherently distinguishable, the number of informative hard samples is limited, and the benefit of weight adjustment is correspondingly smaller.

For basic metric learning methods, they generally yield smaller gains in the full-sample landslide detection scenario. The contrastive learning mechanisms of such methods are primarily designed to address the sparse feature distribution under few-shot settings, where the interclass discriminability and intraclass compactness of the embedding space are the core limiting factors of detection performance. Under the full-sample training paradigm, however, the baseline detector has already learned a relatively mature feature manifold from abundant training data. Basic contrastive constraints with fixed weights and fixed margins provide limited fine-grained reshaping of the embedding space, and the marginal benefit of optimization diminishes, resulting in limited overall performance gains.

In contrast, the proposed OSG-UAHNM avoids the performance dilution of conventional metric learning under full-sample training and maintains stable optimization gains with mild attenuation even in high-contrast scenarios. On the one hand, unlike the fixed contrastive constraints of existing metric learning methods, our proposed UASCL loss adaptively regularizes the embedding space by integrating classification uncertainty for positive samples, joint classification-localization uncertainty for hard negatives, and batch-aware dynamic margins. It simultaneously tightens intraclass feature aggregation and enlarges interclass feature margins, providing fine-grained improvement in feature separability rather than fixed-distance optimization. On the other hand, the dynamic uncertainty-aware mining mechanism adaptively selects the most informative hard samples for training, instead of merely focusing on extreme hard samples with maximum loss, aligning training weights with sample informativeness. This dual optimization mechanism enables the method to sustain stable and accurate detection performance across varying geological and visual contrast conditions, supporting large-scale, rapid landslide screening.

F. Hyperparameter Analysis Experiment

To determine the optimal integration of our metric learning framework with different detector architectures, we conducted a systematic hyperparameter analysis focusing on the contrastive loss weight δ. This analysis was performed using only the OSG module (without UAHNM) to establish the optimal balance between detection loss and metric learning loss for each architecture.

For other hyperparameters, we set $\alpha = \beta = 1.0$ (controlling hard negative and hard positive emphasis), base margin $m_0 = 1.0$, and scaling parameters $\gamma 1 = 1.0$, $\gamma 2 = 0.5$. Given that confidence scores μ_s typically range within [0.5, 0.9] during training and IoU standard deviation σ_{IOU} generally stays below 0.4, these settings ensure the AM m remains within [1.1, 1.9], aligning well with the cosine distance space. In addition, the buffer parameter ϵ in the OSG module is specified as a uniform physical distance of 8 meters rather than a fixed number of pixels. Specifically, this corresponds to $\epsilon = 4$ pixels for the Xiji dataset (2 m/pixel) and $\epsilon = 10$ pixels for the Bijie dataset (0.8 m/pixel). This 8-meter threshold accommodates uncertainty in annotation boundaries while preserving sufficient valid background space for sampling.

As shown in Table XII, optimal δ decreases with model family: Faster R-CNN peaks at $\delta = 0.20$ (+4.6 percentage points in AP), YOLOv8n and YOLO11n at $\delta = 0.10$ (+4.7 and +1.2 percentage points), while YOLO26n and DETR peak at $\delta = 0.05$ (+3.0 and +2.8 percentage points). This trend suggests that feature extractors with stronger baseline representations require weaker contrastive regularization to avoid disrupting their existing representations. Based on this validation set analysis, the optimal δ for each detector is determined and adopted for all subsequent experiments, including ablation study, few-shot learning evaluation, final test set evaluation on both loess and Bijie landslide datasets, and scale-wise performance verification on the Bijie dataset.

V. DISCUSSION

A. Ablation Experiment

Having established the optimal contrastive loss weight δ for each detector architecture in Section IV-F, we now conduct a comprehensive ablation study to quantify the individual contributions of the OSG and UAHNM modules. Specifically, we examine whether the UAHNM module can provide further improvements after the OSG module's hyperparameters have been optimized to achieve peak performance.

Fig. 9 presents the ablation results on the validation set, comparing three configurations for each model: baseline without any enhancement, baseline with OSG using optimal δ, and baseline with both OSG and UAHNM using the same optimal δ. The results show the complementary nature of the two modules.

TABLE XII
OPTIMAL δ VALUES AND PERFORMANCE ACROSS DIFFERENT MODELS ON THE LOESS LANDSLIDE DATASET (VALIDATION SET, OSG ONLY)

Model	Optimal δ	Best AP(%)	AP_{50} (%)	F1 (%)	δ = [0, 0.05, 0.10, 0.15, 0.20, 0.25] (AP%)
Faster R-CNN	0.20	42.4	95.8	93.6	[37.8, 40.6, 40.5, 42.1, 42.4, 40.5]
YOLOv8n	0.10	56.4	97.2	93.2	[51.7, 53.7, 56.4, 52.6, 54.7, 56.1]
YOLO11n	0.10	58.0	98.2	93.1	[56.8, 55.8, 58.0, 56.9, 55.4, 57.6]
YOLO26n	0.05	60.9	95.6	92.4	[57.9, 60.9, 59.0, 57.5, 58.5, 58.0]
DETR	0.05	56.7	94.0	-	[53.9, 56.7, 53.7, 52.6, 53.0, 54.8]

TABLE XIII
COMPONENT-WISE ABLATION OF UASCL ADAPTIVE MECHANISMS ON THE LOESS LANDSLIDE DATASET (VALIDATION SET)

Model	Baseline AP(%)	Fixed AP(%)	+HPW AP(%)	+HNW AP(%)	+AM AP(%)	Full AP(%)
Faster R-CNN	37.8	41.8	41.9	42.0	42.4	42.7
YOLOv8n	51.7	55.6	55.7	55.7	56.0	56.5
YOLO11n	56.8	57.5	57.5	57.6	58.0	58.2
YOLO26n	57.9	59.6	59.8	59.9	60.3	61.0
DETR	53.9	56.9	57.2	57.7	57.5	58.4

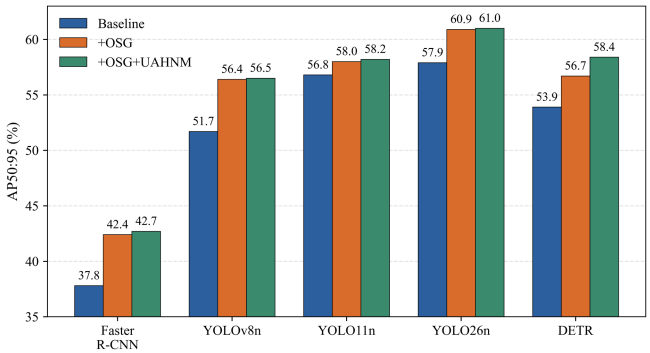


Fig. 9. Ablation study of the proposed method on the loess landslide dataset (validation set).

The OSG module contributed the majority of the overall performance gain, with AP gains of 4.6, 4.7, 2.8, 1.2, and 3.0 percentage points for Faster R-CNN, YOLOv8n, DETR, YOLO11n, and YOLO26n, respectively. These gains indicate that the online sample generation strategy enriches the embedding space by dynamically constructing instance-level contrastive pairs, enabling the model to learn more discriminative representations even under limited supervision.

The subsequent addition of the UAHNM module yielded further improvements beyond the already-optimized OSG configurations, with additional gains of 0.3, 0.1, 1.7, 0.2, and 0.1 percentage points across the respective models. While these incremental gains appear modest for CNN-based detectors, DETR achieved a substantial AP gain of 1.7 percentage points—the largest improvement among all tested models. This larger effect may be associated with DETR's unique characteristics: as a Transformer-based detector with global attention mechanisms, DETR tends to generate a higher proportion of mid-confidence predictions corresponding to ambiguous boundary regions. The UAHNM module's mechanism—mining high-quality hard negatives based on joint localization (IoU) and classification uncertainty scores, and weighting them in the contrastive loss—targets this limitation by explicitly enhancing the model's discriminative capability for challenging, ambiguous samples.

To examine the internal adaptive mechanisms of the proposed UASCL loss and assess the necessity and individual contribution of each component, we conduct an additional component-wise ablation study using the full OSG-UAHNM framework. As summarized in Table XIII, we first define a fixed baseline variant ("Fixed") by removing both the hard positive weighting term $1 + \beta \cdot u_a^{\text{cls}}$ and hard negative weighting term $1 + \alpha \cdot u_a$ from UASCL and fixing the contrastive margin m at 1.0. This configuration retains the core instance-level metric-learning pipeline of OSG-UAHNM but excludes all adaptive weighting and dynamic-margin mechanisms. We then activate each of the three adaptive components—hard positive weighting (HPW), hard negative weighting (HNW), and AM—separately to quantify its individual contribution relative to the fixed variant, and finally evaluate the complete UASCL formulation ("Full").

Among the three components, the AM makes the largest contribution, yielding an average AP gain of 0.56 percentage points over the Fixed configuration. This greater contribution arises from the varying difficulty of contrastive pairs constructed online within each batch. A static margin underpenalizes easily separable negatives, which cease contributing to optimization once they exceed the threshold, while overpenalizing ambiguous samples, which incur excessive loss and destabilize training.

The batch-aware AM addresses this limitation by calibrating the separation threshold to the current sample distribution, thereby enabling more appropriate gradient updates and more stable embedding optimization.

HNW, which complements the UAHNM hard negative miner, yields an average AP gain of 0.30 percentage points over the Fixed variant. UAHNM selects high-quality hard negatives from the predictions, whereas HNW assigns these samples uncertainty-adaptive weights in the loss function. Together, the two mechanisms amplify optimization signals for highly ambiguous samples and form a closed "mining-then-weighting" loop. HNW yields the largest gain for DETR (0.80 percentage points), consistent with the configuration in Table III. DETR retains raw decoder queries without NMS filtering, thereby preserving more ambiguous boundary predictions as hard negative candidates and allowing the weighting mechanism to operate more fully. By contrast, NMS pre-filtering in CNN-based detectors leaves fewer highly ambiguous candidates, resulting in smaller but consistently positive HNW gains.

HPW yields a smaller average gain of 0.14 percentage points over the Fixed variant because well-annotated landslide instances with high intraclass consistency provide limited scope for improvement through positive reweighting. Although the three components make different individual contributions, they act synergistically: the Full configuration achieves an average additional AP gain of 1.08 percentage points over the Fixed baseline. Overall, these adaptive mechanisms provide complementary, fine-grained refinements within UASCL that further sharpen the embedding space, building on the core instance-level metric-learning framework of OSG-UAHNM.

B. Few-Shot Learning Evaluation

To assess the discriminative representation learning capability of OSG-UAHNM under severe label scarcity, we conduct few-shot experiments by reducing the training set to 5-shot, 10-shot, and 15-shot per class, using the optimal contrastive loss weights determined from the hyperparameter analysis in Section IV-F.

As shown in Fig. 10, OSG-UAHNM improves performance in most cases, especially under the 5-shot and 10-shot regimes. Notable gains are observed on Faster R-CNN (+1.84 percentage points at 5-shot), YOLO11n (+2.75 to +11.49 percentage points), YOLO26n (+1.80 to +3.30 percentage points), and DETR (+4.70 to +7.40 percentage points). These results demonstrate that the proposed plug-and-play metric-learning augmentation remains effective when training data are scarce, helping models learn more discriminative instance-level representations from limited supervision.

An exception is YOLOv8n, which underperforms the baseline in most few-shot settings. This result may reflect the fact that YOLOv8n's anchor-free design and refined loss functions already provide strong data efficiency in low-data regimes, leaving less room for contrastive augmentation to contribute. Overall, the framework remains effective in severe few-shot settings across diverse architectures, supporting its applicability to label-scarce remote sensing applications.

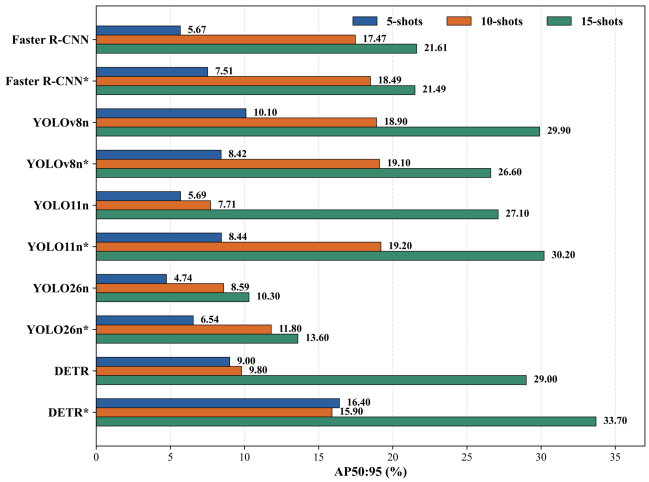


Fig. 10. Few-shot learning evaluation on the loess landslide dataset (test set).

C. Analysis of Intersample Distances in Embedding Space

To characterize the embedding-space changes associated with the performance gains, we track intersample distance evolution in the embedding space throughout training. Fig. 11 compares Euclidean and cosine distance metrics over training epochs, showing distinct behavior.

The Euclidean distance exhibits pronounced oscillations throughout training, particularly for negative pairs. This instability arises from its scale-variant nature in high-dimensional spaces. In contrast, cosine distance demonstrates stable convergence with clear separation patterns. Initially, background negatives and hard negatives are 2.6× and 1.6× farther from the positive anchors, respectively, than are positive pairs. After training with OSG-UAHNM, background negatives and hard negatives reach distances 134× and 102× those of positive pairs, respectively—a two-order-of-magnitude improvement in interclass separation while preserving intraclass compactness.

Cosine distance is also better suited for high-dimensional embeddings. By focusing on angular separation rather than magnitude, cosine distance provides semantically meaningful discrimination, which is relevant to this task where landslide and background terrain exhibit high spectral similarity. The observed distance evolution is consistent with the metric-learning objective: positive pairs are pulled together, background negatives are strongly repelled, and hard negatives are maintained at informative yet challenging distances, establishing a discriminative embedding manifold consistent with representation learning principles [44], [45].

D. Feature Activation Visualization Via Eigen-CAM

To visually examine the changes in feature discrimination associated with OSG-UAHNM, we use Eigen-CAM [46] to compare spatial activation patterns of baseline detectors and their enhanced counterparts.

Fig. 12 shows representative CAM heatmaps across different challenging cases, where warmer colors indicate higher activation intensity. The baseline models tend to produce diffuse

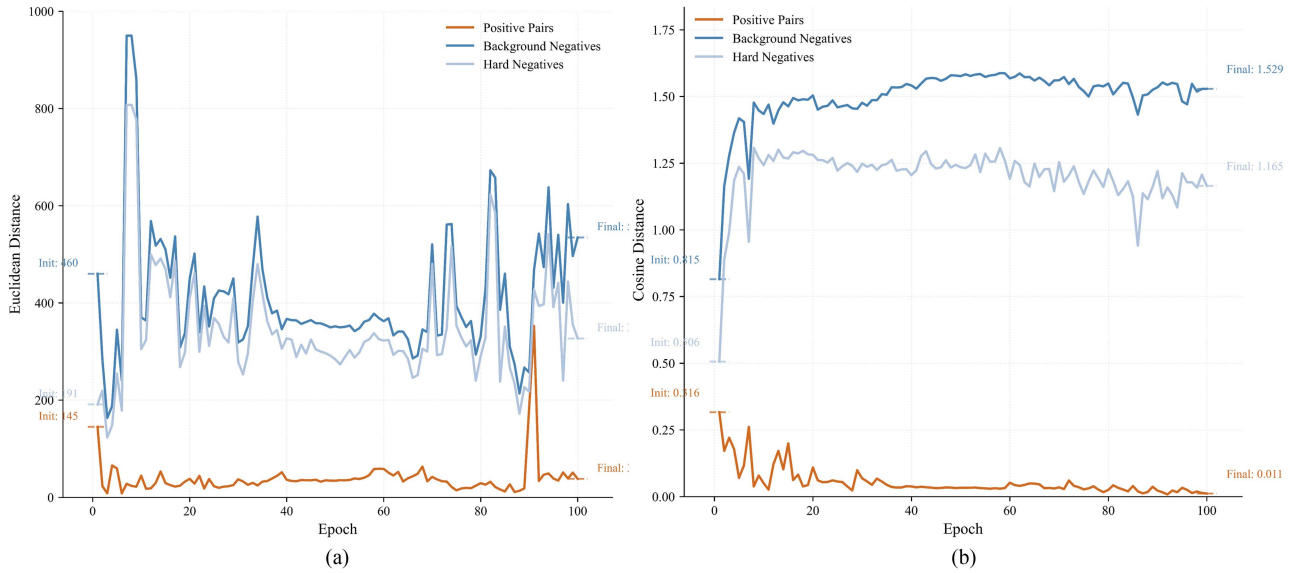


Fig. 11. Evolution of contrastive distances during training. (a) Euclidean distances exhibit significant oscillations due to scale variance. (b) Cosine distances show stable convergence, with background negatives and hard negatives reaching 134× and 102× the distance of positive pairs respectively (from initial 2.6× and 1.6×), demonstrating the framework's effectiveness in establishing discriminative embedding geometry. All distances are computed using features extracted by ResNet-101 backbone.

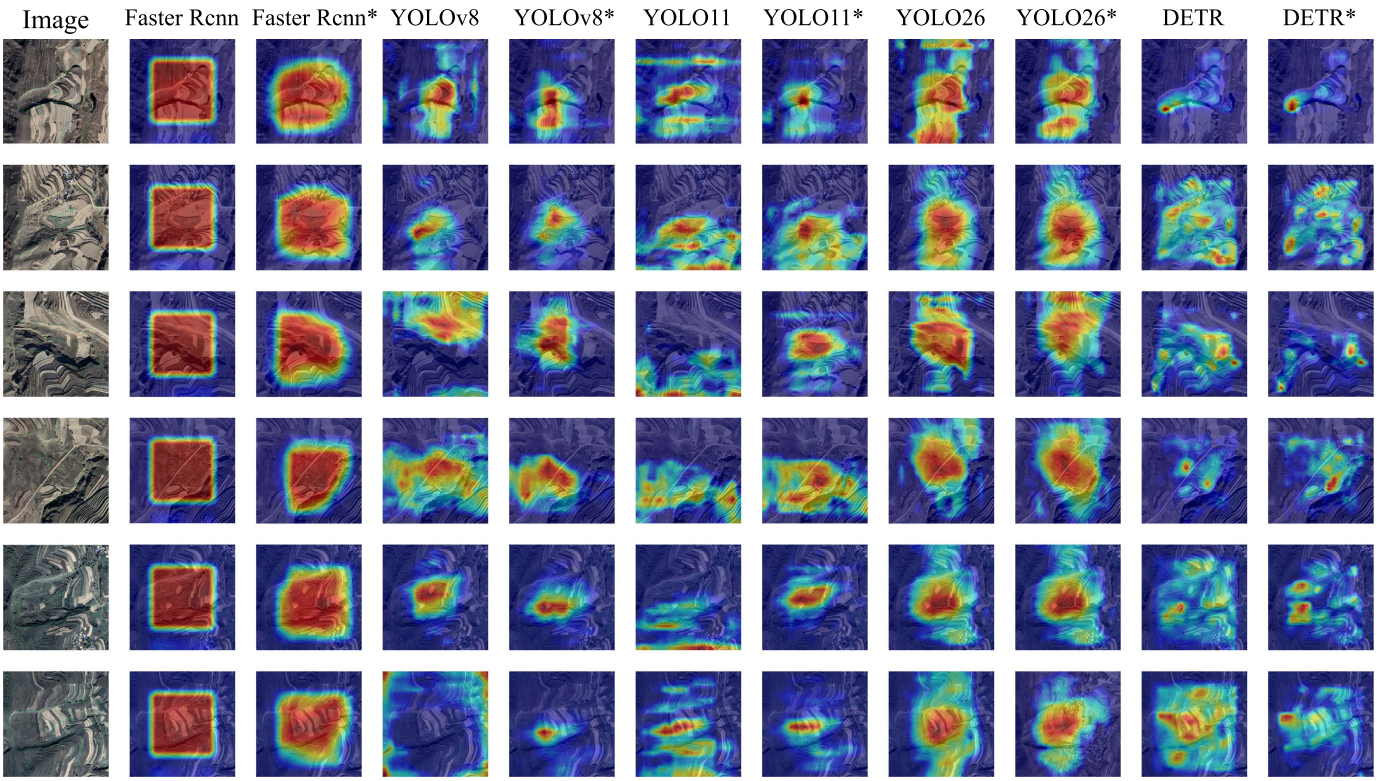


Fig. 12. Eigen-CAM visualization comparing baseline and enhanced detectors. Heatmaps show spatial activation patterns (warmer colors = higher intensity). Enhanced models (∗) produce sharper, more focused activations on landslide-discriminative regions while suppressing background responses, demonstrating improved feature discriminability across all architectures.

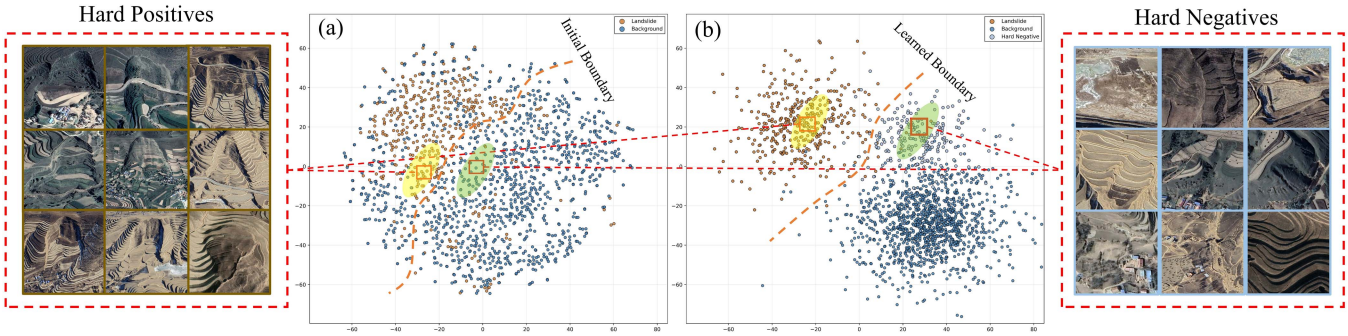


Fig. 13. t-SNE visualization of embedding space evolution. (a) Initialization shows heavily intertwined landslide (brown) and background (dark blue) samples. Hard positive-negative pairs near ambiguous separation boundaries yield an average cosine distance of 0.506. (b) After 100 epochs of OSG-UAHNM training, three distinct clusters emerge: landslides, backgrounds, and hard negatives (light blue) distributed alongside refined separation boundaries, where the average distance of hard positive-negative pairs rises to 1.165. Representative patches illustrate sample characteristics at boxed locations.

activations, often spreading to background regions or missing key landslide boundaries. In contrast, our enhanced models (∗) generate sharper and more focused responses aligned with landslide scarps, boundaries, and debris fans. This improvement is consistent across all detectors: the enhanced versions maintain coherent coverage over ambiguous loess landslide boundaries and effectively suppress false activations on background features like erosion gullies. These patterns indicate that the framework improves feature discriminability, enabling detectors to locate true landslide instances with higher precision regardless of the underlying architecture.

E. Embedding Quality Analysis

To assess the effectiveness of the proposed OSG-UAHNM contrastive learning framework, we visualize the feature embedding distributions before and after training using t-SNE [47] in Fig. 13. In the figure, brown points denote landslide instances (positive samples), dark blue points represent background patches, and light blue points indicate hard negatives.

As shown in Fig. 13(a), the initial feature distribution is extracted from the ImageNet-pretrained Res-Net101 backbone. After training with the OSG-UAHNM framework [see Fig. 13(b)], the features form compact and well-separated clusters, with hard negatives located at the decision boundary.

F. Geological Context and Material-Induced Detection Challenges

While the proposed framework exhibits consistent generalization across datasets, the difficulty of the detection task is influenced by the material properties and geological setting of the study area, which were explicitly considered during dataset construction. Xiji County lies within the meizoseismal zone of the 1920 M8.5 Haiyuan earthquake, in a loess hilly–gully landscape sculpted by seismic faulting and prolonged erosion [44]. Loess is a structurally metastable, water-sensitive deposit characterized by high porosity, developed vertical joints, and strong collapsibility upon wetting [44]. These attributes predispose the region to slope failures that frequently evolve into high-mobility flowslides once the saturated material is fluidized [45]. Accordingly, the imagery was not treated as a purely visual source during annotation: each landslide delineation was cross verified against field-survey observations and UAV-derived photogrammetric models to ensure that the labels captured the true geomorphic extent of flow-like deposition zones.

Compared with landslides developed in other geological settings, such as the mountainous rock and soil slides represented by the Bijie dataset [48], the loess landslides in the study area pose specific challenges for object detection. First, unlike mountainous landslides that often expose distinct main scarps and fresh bedrock [51], the displaced loess material shares nearly identical spectral reflectance and textural patterns with the surrounding undisturbed loess landforms, yielding intrinsically low target–background contrast.

Second, owing to the fluid-like rheology of saturated loess during failure, loess flowslides accumulate in fan- or tongue shaped deposition zones with diffuse, nonrigid boundaries [45], which provides convolutional feature extractors with limited edge information. Finally, loess deposits are rapidly recolonized by vegetation during post-failure recovery, and the resulting canopy is spectrally similar to that of adjacent natural slopes, progressively erasing residual visual traces [3], [4].

These material- and setting-induced factors make loess landslides a challenging testbed for discriminative representation learning. The spectral confusion and diffuse boundaries imply that the training data contain background regions that are visually indistinguishable from genuine landslide material, that is, geologically induced hard negatives. Detectors relying predominantly on superficial edge cues can saturate under such conditions. Our UAHNM explicitly targets these geologically induced ambiguous samples, encouraging the network to encode subtle, high-level geomorphic evidence rather than superficial contrast. This provides a process-based rationale for the effectiveness of the proposed instance-level deep metric learning approach.

VI. Conclusion

In this article, we address the challenge of loess landslide detection in settings where low visual contrast and expert annotations remain scarce. Rather than increasing architectural

complexity, we focus on how models learn discriminative representations under limited supervision.

We propose OSG-UAHNM, a plug-and-play deep metric learning framework that formulates landslide detection as an embedding-space optimization problem. The framework operates entirely through loss augmentation, with negligible parameter overhead and zero inference cost, albeit with noticeable additional training computation. It comprises two complementary components. The OSG dynamically constructs instance-level contrastive pairs using labeled landslides and unlabeled backgrounds, with buffered zero-IoU sampling to exclude potential landslide marginal features from background patches. The UAHNM identifies informative samples by jointly modeling classification confidence and localization precision. Together, they establish a structured embedding manifold where interclass separation increases by over two orders of magnitude while maintaining intraclass compactness.

Experiments show consistent improvements across diverse architectures: Faster R-CNN (+7.6 percentage points in AP), YOLO series (+2.7 to +6.5 percentage points), and DETR (+1.8 percentage points). Parallel evaluations on the Bijie dataset yield consistent gains that support cross-dataset robustness, and AP gains are also observed when metrics are stratified by landslide size. We also compare our framework with existing metric learning and hard example mining methods designed for object detection, and our approach achieves more stable performance across scenes with varying visual contrast. Feature activation visualizations show sharper attention on landslide regions with reduced background confusion, providing interpretable evidence that learned representations improve discriminative capability. Furthermore, few-shot learning validation confirms that our approach still delivers performance gains under severe annotation scarcity, outperforming baseline methods across most models in data-scarce scenarios. By seamlessly integrating into standard detection pipelines, this work provides a readily deployable approach for learning discriminative representations in label-scarce remote sensing applications where targets exhibit low visual contrast, as well as general object detection scenarios.

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